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ATTIC VENTILATION DEFECT REPORT

Property Address: 1517 San Juan Avenue Victoria, BC V8N 2L4	**Report Date:** January 29, 2026 **Property Owner:** Richard Fremmerlid **Prepared By:** Justin Curran, EIT

1. Executive Summary

A site inspection of the attic ventilation system at 1517 San Juan Avenue revealed significant deficiencies in the current roof ventilation configuration. The installed Menzies SV-50 roof vents have been incorrectly specified and installed as intake vents on the lower roof slope, contrary to manufacturer specifications and building science principles. This has resulted in a ventilation imbalance creating negative pressure conditions, moisture accumulation, and early signs of mold growth on the roof sheathing.

2. Defects Observed

2.1 Ventilation Imbalance

The current ventilation configuration shows a significant imbalance between exhaust and intake capacity. Measured ventilation areas indicate approximately 1,080 square inches of exhaust ventilation (from 12 SV-50 vents at 50 sq. in. each, plus existing ridge or upper venting) versus approximately 700 square inches of intake ventilation. This represents a 35% deficit in intake capacity relative to exhaust. Building science research consistently establishes that intake ventilation should equal or exceed exhaust ventilation, with an ideal ratio of 50-60% intake to 40-50% exhaust.

2.2 Improper Vent Placement and Specification

The twelve Menzies SV-50 vents are positioned on the lower roof slope rather than at the soffits or eave level where intake ventilation is required. The manufacturer explicitly states that the SV-50 is an "Exhaust vent for mounting at upper portion of steep roof. Not to be used as an intake vent." The original roofing specification incorrectly designated these units as "air intake vents," which contradicts both manufacturer guidelines and proper ventilation design principles.

2.3 Insufficient Soffit Intake

The property currently has only four small soffit vents on the east and west elevations, with no soffit intake ventilation on the north and south elevations. This configuration fails to provide adequate intake air at the lowest point of the attic space as required for proper cross-ventilation. The absence of continuous soffit ventilation creates dead zones where air cannot circulate, leading to moisture accumulation and potential structural damage.

2.4 Moisture and Mold Conditions

The improper ventilation configuration has created a "vacuum effect" where exhaust vents draw conditioned air from the living space through ceiling penetrations rather than drawing fresh outdoor air through proper intake vents. This introduces warm, moist interior air into the attic space, where it condenses on the cooler roof sheathing. Evidence of moisture damage and early mold growth was observed during the inspection, indicating that remediation is required to prevent further deterioration of the roof structure.

3. Building Code and Building Science Justification

3.1 BC Building Code 9.19.1.3

The British Columbia Building Code Section 9.19.1.3 requires that "not less than 63 mm of space shall be provided between the top of the insulation and the underside of the roof sheathing." Furthermore, at the junction of sloped roofs and exterior walls, the code requires that ventilation baffles "provide an unobstructed air space" that is "of sufficient cross area to meet the attic or roof space venting requirements." The current SV-50 configuration does not provide intake ventilation at the eave level, creating a dead zone that prevents air circulation at this critical junction point.

3.2 Manufacturer Specifications (Menzies SV-50)

The Menzies Metal Products specification sheet for the SV-50 Plastic Roof Vent explicitly states: "Exhaust vent for mounting at upper portion of steep roof (minimum pitch 3.5:12). Not to be used as an intake vent." The SV-50 is designed and tested as an exhaust device, with its geometry optimized to prevent rain and snow infiltration when installed at upper roof positions. Using these vents as intake devices on lower roof slopes contradicts the product's intended application and violates the conditions under which it is certified to CSA CAN3-A93 and IAPMO PS 64 standards.

3.3 Building Science Principles

Recognized building science authorities, including Dr. Joseph Lstiburek of Building Science Corporation, establish that effective attic ventilation requires intake at the lowest point (soffits/eaves) and exhaust at the highest point (ridge). The principle is to "wash the entire underside of the roof deck with cold air," allowing buoyancy-driven airflow to carry moisture out of the attic space.

Multiple industry sources corroborate these principles:

Source	Key Finding
Building Science Corporation	Recommends 50-75% of ventilation at eaves; 60/40 intake/exhaust ratio is the "sweet spot"
GAF Building Science	40-50% exhaust at ridge, remainder at eaves; "never more exhaust than intake"
Roofing Contractor Magazine	"The number one rule...there should never be more exhaust than intake ventilation"
Asphalt Roofing Manufacturers Association (ARMA)	Imbalanced ventilation causes premature shingle deterioration and moisture damage; voids warranty coverage
Owens Corning Research (AtticSim)	Balanced soffit-to-ridge ventilation reduces attic temperatures by 30°F+ compared to unbalanced systems

The current configuration at 1517 San Juan Avenue creates "short-circuiting" where exhaust vents draw air from the nearby SV-50s rather than from proper soffit intake. This defeats the purpose of ventilation and allows moisture to accumulate in areas that receive no airflow.

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4. Recommendations

Based on the deficiencies documented above, I recommend the following remediation work:

4.1 Remove SV-50 Vents and Seal Openings

All twelve Menzies SV-50 vents should be removed from the lower roof slope. The roof penetrations must be properly sealed with new plywood patches cut to fit, underlayment reinstalled (Ice & Water Shield or equivalent), and shingles woven back into the existing roof surface. This work should include proper flashing and sealant application to ensure a watertight repair.

4.2 Install Proper Soffit Ventilation

Continuous soffit ventilation should be installed at the eave level on all elevations of the home. Three options are available:

Option	Description	Considerations
A	Continuous channel cut into existing stucco soffit (3-4" wide strip)	Functional but visible cut line; lower material cost, moderate labor
B	Circular vent pucks (4" diameter) cut into existing soffit at intervals	Multiple visible circles; lower material cost, moderate labor
C	Full soffit replacement with vented material (vinyl, aluminum, or wood)	Clean, integrated appearance; higher material and labor cost; allows recessed lighting

Option C is recommended based on the homeowner's preference for a clean, modern aesthetic with integrated recessed lighting and natural wood appearance.

4.3 Air-Seal the Attic Floor

To reduce moisture migration from the living space into the attic, air sealing should be performed at all penetrations through the ceiling plane. This includes sealing around electrical boxes, plumbing stacks, HVAC ducts, and the attic hatch. This work may be completed in a subsequent phase if budget constraints require phasing the remediation.

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5. Conclusion

The attic ventilation system at 1517 San Juan Avenue contains significant deficiencies that violate building code requirements, manufacturer specifications, and established building science principles. The Menzies SV-50 vents were incorrectly specified and installed in a manner that contradicts their intended use as exhaust-only devices. Remediation is required to prevent further moisture damage to the roof structure.

This report is provided to document the observed deficiencies and may be used to support a warranty claim with the original roofing contractor regarding the improper specification and installation of the SV-50 vents.

Respectfully submitted,

Justin Curran
Chief Executive Officer
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